

Partial Derivatives

Multi-input function mein ek variable ka effect, baaki fix rakh kar

01 Opening Hook

STORY TIME

Socho tum ek recipe bana rahe ho jisme namak aur mirch dono add hote hain taste par. Agar tumhe jaanna hai 'sirf namak badhane se taste kitna change hoga', tum mirch ko constant rakh kar sirf namak change karte ho. Yeh hi hai **Partial Derivative**.

02 Why Does This Concept Exist?

THE PROBLEM

Neural networks mein ek function (loss) ke hazaron/millions inputs (weights) hote hain. Humein jaanna hota hai har individual weight ka loss par kya effect hai, baaki sab constant rakh kar. Partial derivative invent hua taaki hum multi-variable functions mein ek variable ka isolated effect measure kar sakein.

03 Explain Like I'm 10

SIMPLE WORDS

Socho tumhare pizza ka taste 2 cheezon par depend karta hai - cheese amount aur sauce amount. Agar tum poochte ho 'sirf cheese badhane se taste kitna better hoga (sauce same rakh kar)', yeh sawal partial derivative hai cheese ke respect mein.

04 Build Intuition First (No Maths Yet)

FEEL IT FIRST

Partial derivative ek multi-variable function mein ek variable ko change karne ka effect measure karta hai, baaki saare variables ko 'frozen' (constant) rakh kar. Neural network ke saare weights ke liye alag-alag partial derivatives calculate kiye jaate hain.

05 Ab Thoda Maths

FORMULA TIME

Notation:

$\partial f / \partial x$ - yeh function f ka partial derivative x ke respect mein hai, baaki variables constant.

Example:

Agar $f(x,y) = x^2 + 3xy$, toh $\partial f / \partial x = 2x + 3y$ (y ko constant maan kar), aur $\partial f / \partial y = 3x$ (x ko constant maan kar).

06 Visualize Everything

PICTURE IT

$f(x, y) = x^2 + y^2$ (ek bowl-shaped surface)

$\partial f / \partial x$ = sirf x -direction mein slope

$\partial f / \partial y$ = sirf y -direction mein slope

07 AI Connection

WHERE AI USES THIS

Gradient Vector: Neural network ke loss function ke saare partial derivatives (har weight ke respect mein) mila kar 'gradient vector' banta hai.

Backpropagation: Backprop algorithm har layer ke weights ke partial derivatives calculate karta hai chain rule use karke.

08 Real Industry Example

REAL COMPANIES

All AI companies: Training ke har step mein millions of partial derivatives (ek har parameter ke liye) calculate hote hain automatically.

09 Explain Like I'm Going To Build It

HOW YOU'D CODE IT

PyTorch mein jab tum multi-parameter model par `.backward()` call karte ho, PyTorch automatically har parameter ka partial derivative calculate karke `.grad` attribute mein store kar deta hai.

10 Common Misconceptions

BEGINNER MISTAKES

Galat soch: Partial derivative pura function ka behavior batata hai.

Sahi baat: Partial derivative sirf ek direction (ek variable) mein change batata hai - poora picture (gradient vector) saare partial derivatives ko combine karke milta hai.

11 Interview Questions

INTERVIEW CORNER

[Beginner] Q: Partial derivative kya hai?

A: Multi-variable function mein ek variable ka effect, baaki variables ko constant rakh kar measure karna.

[Intermediate] Q: Neural network mein partial derivatives kyun zaroori hain?

A: Kyunki loss function millions of weights par depend karta hai - humein har weight ka individual contribution jaanna zaroori hai use sahi tarique se update karne ke liye.

[Advanced] Q: Gradient vector aur partial derivatives mein kya relation hai?

A: Gradient vector saare partial derivatives (har variable ke respect mein) ka collection hai - yeh function ke sabse steep increase ki direction batata hai multi-dimensional space mein.

12 Exercises

PRACTICE TIME

[Easy] $f(x,y) = x^2 + y^2$, iska $\partial f / \partial x$ kya hoga?

Solution: $2x$.

[Medium] $f(x,y) = 3x + 2xy$, iska $\partial f / \partial y$ kya hoga?

Solution: $2x$ (kyunki $3x$ constant hai y ke respect mein, aur $2xy$ ka derivative $2x$ hai).

[Hard] Neural network mein agar ek weight ka partial derivative bahut chhota (near-zero) hai, iska training par kya effect hoga?

Solution: Woh weight bahut slowly update hoga (vanishing gradient jaisa effect), jisse us particular connection ka learning bahut slow ho jaayega.

13 Golden Summary

Concept	Easy Meaning
Partial Derivative	Ek variable ka effect, baaki constant rakh kar
Notation	$\partial f / \partial x$
Gradient Vector	Saare partial derivatives ka collection
Neural Network Use	Har weight ka individual contribution measure karna
Backprop Role	Layer-by-layer partial derivatives calculate karna

14 One-Line Memory Trick

“Partial derivative sirf ek ingredient badalta hai, baaki sab same rakh kar.”